A Cross-City Spatial Machine Learning Framework for Mapping Urban Heat Risk

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Background

Urban heat is a growing public health, infrastructure, and environmental justice issue because temperatures can vary sharply within the same city over short distances [1]. These differences are shaped by characteristics such as impervious cover, vegetation, topography, and proximity to water, which influence how urban surfaces absorb, store, and release heat [2]. As a result, nearby neighborhoods can experience different levels of heat exposure under the same local weather conditions. To study these patterns, this project builds a geospatial dataset for 30 U.S. cities representing hot/arid, hot/humid, and mild/cool climates. Our goal is to create a cross-city dataset to examine how land-surface characteristics affect urban heat risk.

Our dataset combines several publicly available remote-sensing and mapping products from U.S. government agencies. Two key predictors, land cover and impervious surface, come from the National Land Cover Database (NLCD), produced by the Multi-Resolution Land Characteristics Consortium [3][4]. NLCD was developed to provide standardized information on land cover and land-cover change across the United States using 30 m satellite imagery [3]. In this project, the "land cover" layer describes the general surface composition of each location, while "impervious surface" measures the share of each cell covered by built materials such as roads, rooftops, and pavement [4]. These variables are central to urban heat analysis because developed surfaces tend to retain more heat and reduce evaporative cooling relative to natural land covers.

Elevation is obtained from the U.S. Geological Survey's 3D Elevation Program (3DEP) [5]. 3DEP was created to provide consistent national elevation data for mapping and environmental analysis, and its digital elevation models are distributed by USGS at resolutions including about 10 m [5]. Elevation is included because terrain can affect airflow, drainage, and localized heat buildup. Distance to water is derived from NHDPlus High Resolution, a USGS hydrography dataset representing streams, rivers, lakes, and related water features [7]. NHDPlus HR was developed to support hydrologic and spatial analysis [7]. Distance to water is a relevant predictor of within-city heat variation because water features influence land surface temperature through cooling and moisture effects. Vegetation is represented using NDVI derived from Landsat Collection 2 Level-2 Surface Reflectance imagery [8]. The Landsat program is jointly operated by NASA and USGS, and its Level-2 products provide atmospherically corrected spectral data suitable for environmental analysis [8]. NDVI, calculated from red and near-infrared reflectance, is widely used as an indicator of vegetation abundance and condition. Vegetation is critical because greener areas are associated with shade and evapotranspirative cooling.

The response variable is land surface temperature (LST) from ECOSTRESS, the ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station [6]. ECOSTRESS is a NASA thermal remote sensing mission mounted on the International Space Station and designed to measure LST at relatively fine spatial resolution [6]. These data are especially useful for urban heat research because they capture localized thermal patterns within cities. In this project, ECOSTRESS provides the temperature outcome used to identify elevated urban heat conditions across cities with different climates.

Summary of Data

Our project dataset contains filtered geospatial features for 30 U.S. cities spanning hot/arid, hot/humid, and mild/cool climate types. The original sources for these features are outlined in the background section. For each city, we project a standardized 30 m grid within a

buffered study area defined as the Census urban area plus a 2,000 m buffer, then align predictors and outcomes to this common geometry. Core variables include median May–August land surface temperature (LST), a binary hotspot indicator for cells at or above the city-specific 90th percentile of LST, impervious surface share, median warm-season NDVI, distance to water, land-cover class, elevation, and the number of valid ECOSTRESS passes.

The remainder of this section will focus on Phoenix, AZ as a preliminary subset of the full 30-city dataset. Our Phoenix dataset contains 4,735,561 filtered grid cells and 14 variables, with each observation representing one 30 m cell; only 7,286 open-water cells and 763 cells with fewer than three valid ECOSTRESS passes were dropped. Missingness is negligible: just 24 NDVI values are missing (0.0005% of observations) and all other key variables are fully observed. As illustrated in Figure 1, median May–August LST is roughly bell-shaped with a mean of 315.09 and a 10th-to-90th percentile range from 312.02 to 317.77, which are reasonable for the area. Predictors show greater heterogeneity. Impervious surface shows a median of 27.79%, with the 10th percentile at 0.00% and the 90th percentile at 67.03%, and is strongly left-skewed. This is likely the result of the study area including many regions of undeveloped desert in addition to urban areas and may warrant a reconsideration of our study area buffer. Distance to water is also strongly skewed, with a mean of 550.68 m, median of 390.00 m, 90th percentile of 1,236.93 m, and skewness of 2.05. A logarithmic transformation is applied for visualization, under which DtW takes a roughly bell-shaped distribution. Elevation also varies substantially, with a mean of 450.07 m and a 10th-to-90th percentile range from 341.42 to 610.23 m. Because all variables are derived from standardized national data sources, we expect data for the remaining 29 cities to be comparable in completeness and cleanliness.

Figure 2 shows that Phoenix land cover is dominated by Shrub/Scrub (30.43%), consistent with Phoenix's location in the Sonoran Desert. Figure 3 presents the correlation matrix for key predictors and median LST and suggests that the strongest simple linear association with LST is elevation ($r = -0.52$), followed by ECOSTRESS pass count ($r = -0.37$) and NDVI ($r = -0.21$), whereas impervious surface ($r = 0.05$) and distance to water ($r = 0.03$) show only weak relationships. Although these weak correlations are slightly surprising, they may be a result of the matrix reflecting only unadjusted linear relationships and obscuring more complex or nonlinear effects. Noting these correlations, we move on to Figure 4, which maps hotspot cells. The highest concentration of hotspots appears in the areas furthest from urban development, likely due to the surrounding desert. Phoenix may present an outlier from other study cities in this regard. Visually, hotspots appear more concentrated along the outer portions of the study area, while the northern portion contains relatively few hotspot cells; this pattern is broadly consistent with the negative association between LST and elevation. Some hotspot clusters also appear near the Salt and Agua Fria Rivers, but because the citywide correlation between LST and distance to water is negligible, these visible patterns should be interpreted cautiously.

Potential Problems

Our 30-city dataset is designed to provide insight into the association between distinct land-surface characteristics and urban heat risk. More specifically, we aim to answer three related questions. First, which characteristics are most strongly associated with higher land surface temperature and hotspot occurrence within cities? Second, do these relationships generalize across cities, or do they differ systematically across hot/arid, hot/humid, and mild/cool climate types? Third, can a model trained on a diverse set of cities accurately identify areas of high heat risk in held-out and new cities using select geospatial predictors?

Tables and Figures

Table 1.

Hot / arid:	**Hot / humid:**	**Mild / cool:**
Phoenix, AZ	Houston, TX	Seattle, WA
Tucson, AZ	Columbia, SC	Portland, OR
Las Vegas, NV	Richmond, VA	San Francisco, CA
Albuquerque, NM	New Orleans, LA	San Jose, CA
El Paso, TX	Tampa, FL	Los Angeles, CA
Denver, CO	Miami, FL	San Diego, CA
Salt Lake City, UT	Jacksonville, FL	Chicago, IL
Fresno, CA	Atlanta, GA	Minneapolis, MN
Bakersfield, CA	Charlotte, NC	Detroit, MI
Reno, NV	Nashville, TN	Boston, MA

Figure 1.

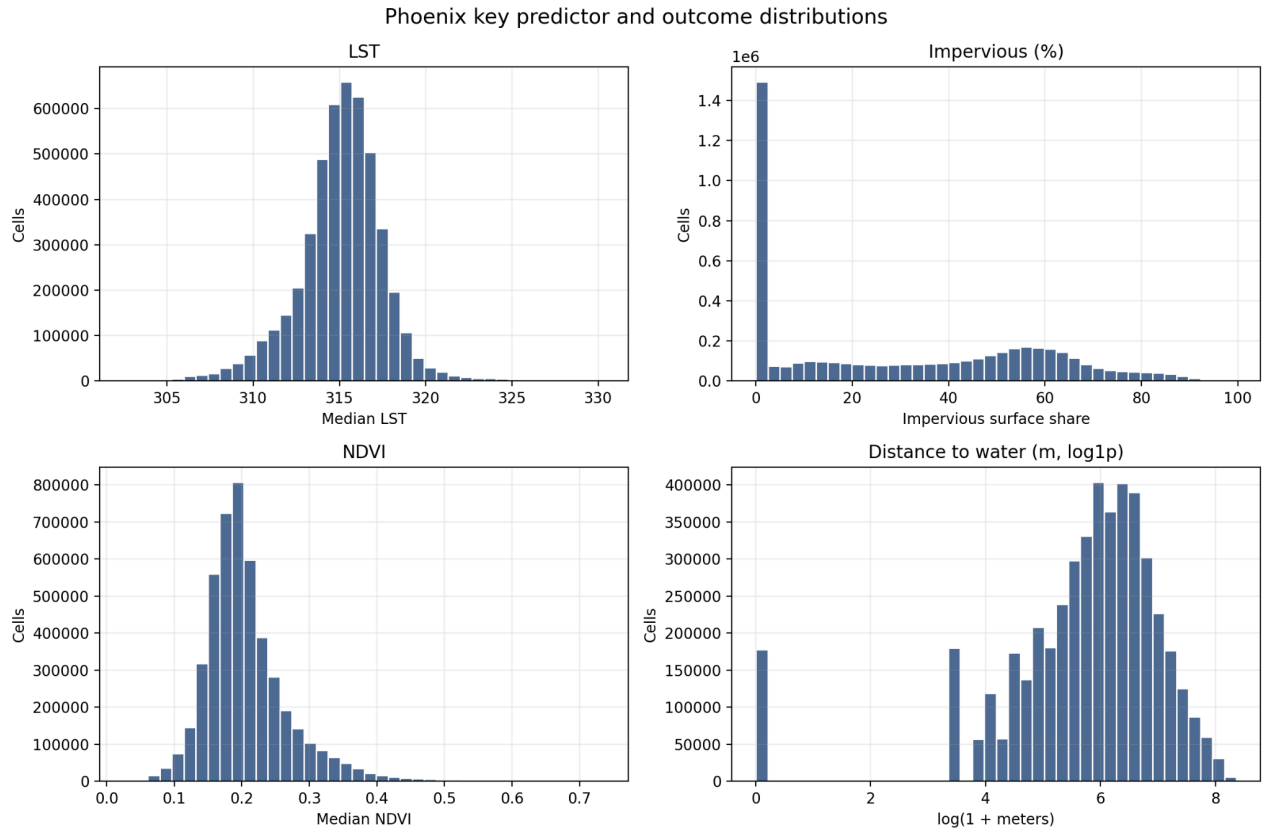

Phoenix key predictor and outcome distributions
LST
Cells
Median LST
Impervious (%)
1e6
Cells
Impervious surface share
NDVI
Cells
Median NDVI
Distance to water (m, log1p)
Cells
log(1 + meters)

Figure 2.

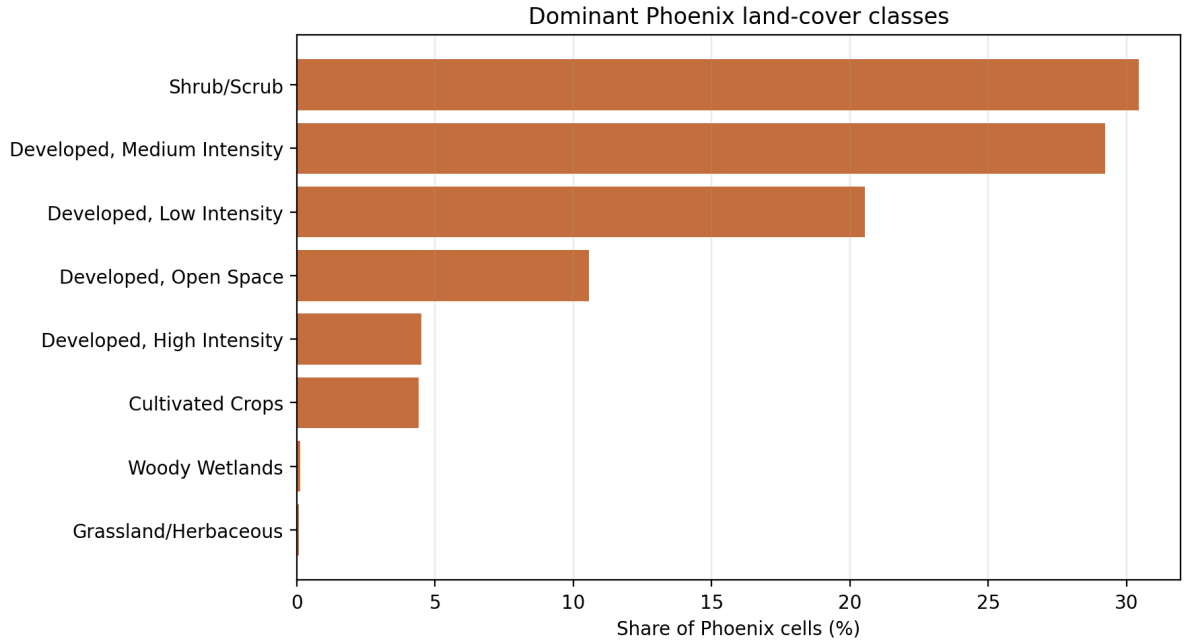

Dominant Phoenix land-cover classes
Shrub/Scrub
Developed, Medium Intensity
Developed, Low Intensity
Developed, Open Space
Developed, High Intensity
Cultivated Crops
Woody Wetlands
Grassland/Herbaceous
0
5
10
15
20
25
30
Share of Phoenix cells (%)

Figure 3.

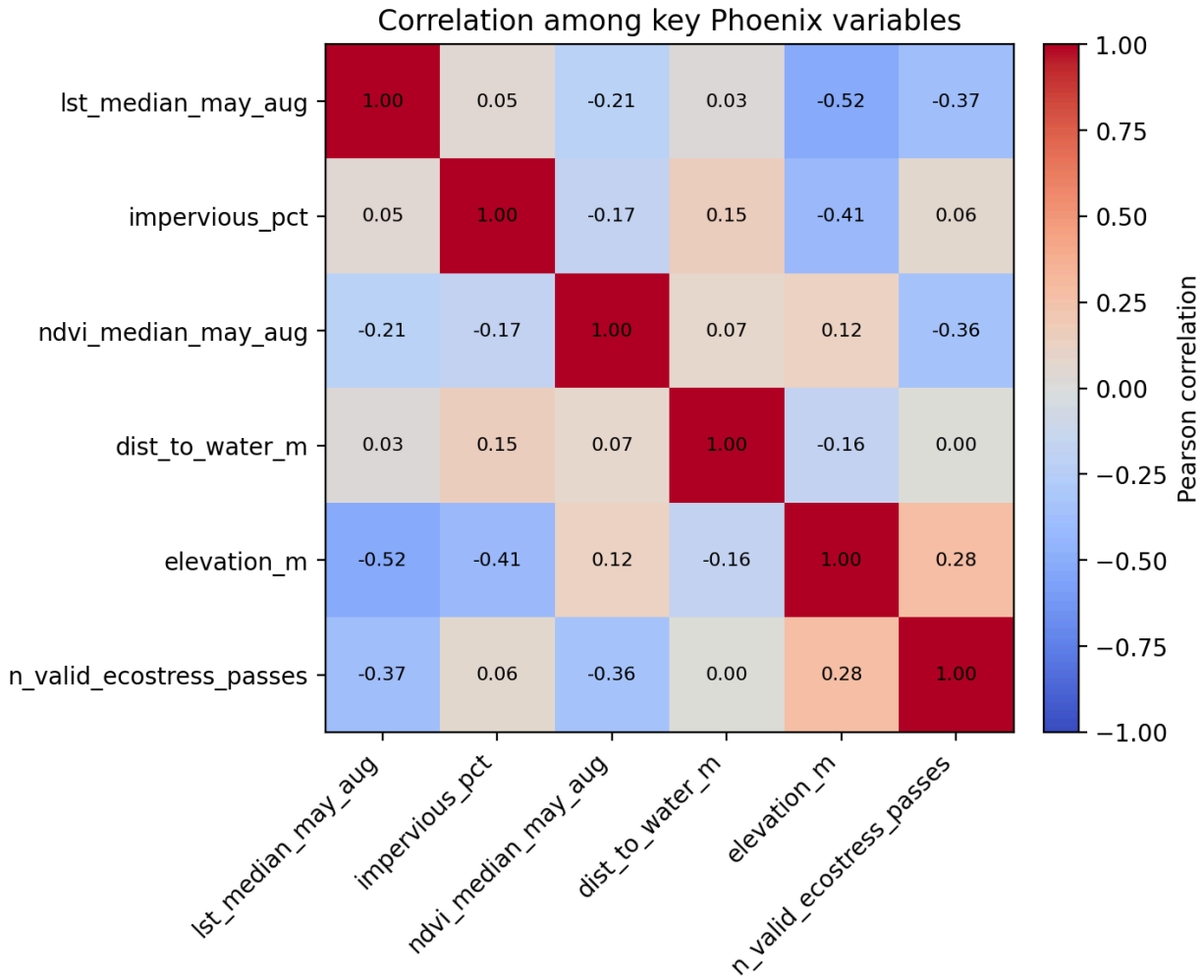
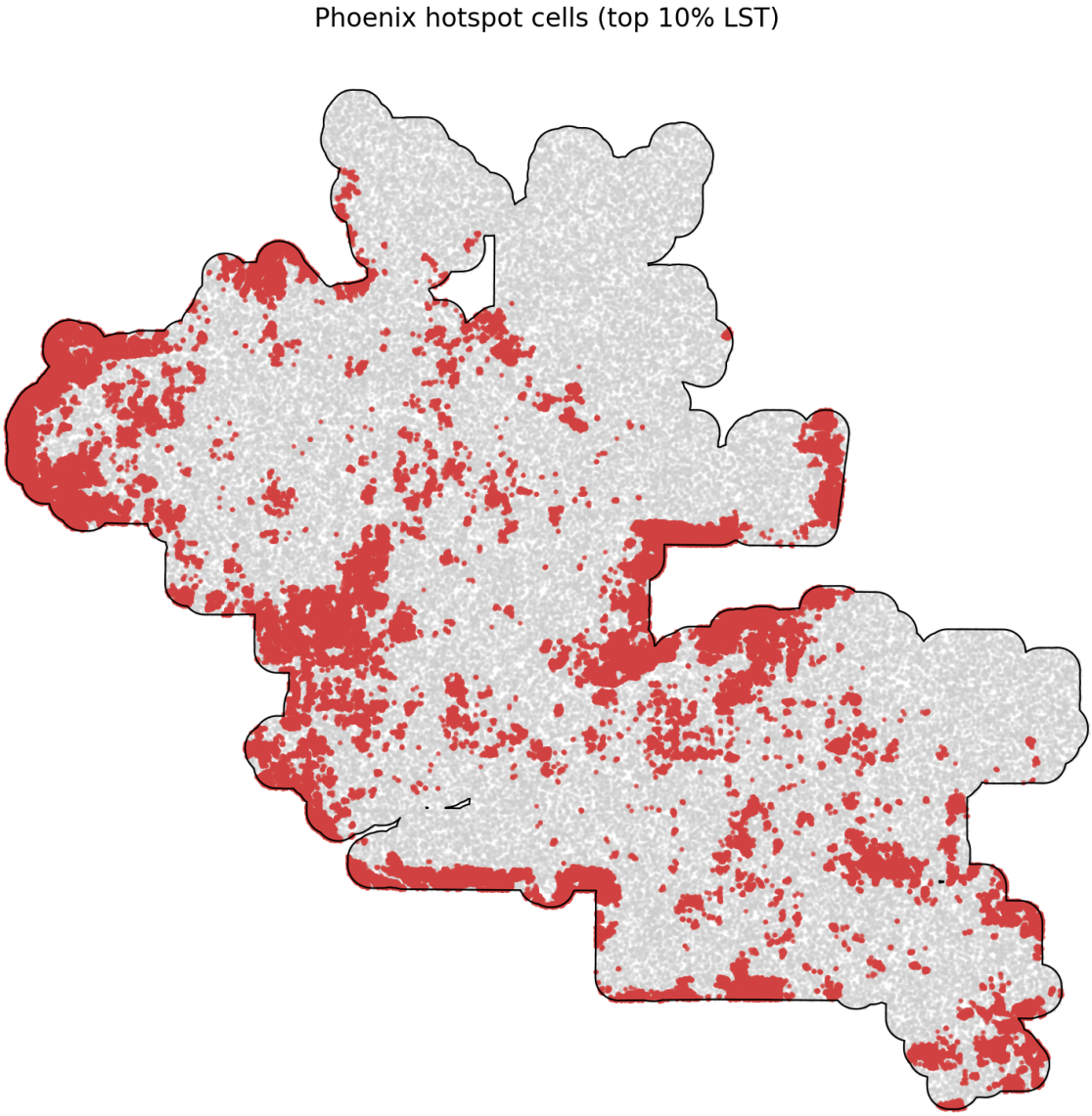

Correlation among key Phoenix variables
lst_median_may_aug
impervious_pct
ndvi_median_may_aug
dist_to_water_m
elevation_m
n_valid_ecostress_passes
1.00 0.05 -0.21 0.03 -0.52 -0.37
0.05 1.00 -0.17 0.15 -0.41 0.06
-0.21 -0.17 1.00 0.07 0.12 -0.36
0.03 0.15 0.07 1.00 -0.16 0.00
-0.52 -0.41 0.12 -0.16 1.00 0.28
-0.37 0.06 -0.36 0.00 0.28 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Pearson correlation

Figure 4.



References

1. U.S. Environmental Protection Agency. "Measuring Heat Islands." Accessed March 7, 2026. https://www.epa.gov/heatislands/measuring-heat-islands.
2. U.S. Environmental Protection Agency. "Compendium of Strategies; Urban Heat Island Basics." Accessed March 7, 2026. https://www.epa.gov/sites/default/files/2014-06/documents/basicscompendium.pdf.
3. Multi-Resolution Land Characteristics Consortium. "Land Cover." Accessed March 7, 2026. https://www.mrlc.gov/data/type/land-cover.
4. Multi-Resolution Land Characteristics Consortium. "Fractional Impervious Surface." Accessed March 7, 2026. https://www.mrlc.gov/data/type/fractional-impervious-surface.
5. U.S. Geological Survey. "About 3DEP Products & Services." Accessed March 7, 2026. https://www.usgs.gov/3d-elevation-program/about-3dep-products-services.
6. NASA Earthdata. "ECOSTRESS Instrument Data." Accessed March 7, 2026. https://www.earthdata.nasa.gov/data/instruments/ecostress.
7. U.S. Geological Survey. "NHDPlus High Resolution." Accessed March 7, 2026. https://www.usgs.gov/national-hydrography/nhdplus-high-resolution.
8. U.S. Geological Survey. "Landsat Collection 2 Surface Reflectance." Accessed March 7, 2026. https://www.usgs.gov/landsat-missions/landsat-collection-2-surface-reflectance.

Appendix A

Code for generating figures.

```
def _save_distribution_figure(df: pd.DataFrame, output_path: Path) -> None:
    fig, axes = plt.subplots(2, 2, figsize=(12, 8))
    plot_specs = [
        ("lst_median_may_aug", "LST", axes[0, 0], None),
        ("impervious_pct", "Impervious (%)", axes[0, 1], None),
        ("ndvi_median_may_aug", "NDVI", axes[1, 0], None),
        ("dist_to_water_m", "Distance to water (m, log1p)", axes[1, 1], np.log1p),
    ]
    for column, title, ax, transform in plot_specs:
        series = pd.to_numeric(df[column], errors="coerce").dropna()
        values = transform(series) if transform is not None else series
        ax.hist(values, bins=40, color="#4C6A92", edgecolor="white")
        ax.set_title(title)
        ax.set_ylabel("Cells")
        ax.grid(alpha=0.2)
    axes[0, 0].set_xlabel("Median LST")
    axes[0, 1].set_xlabel("Impervious surface share")
    axes[1, 0].set_xlabel("Median NDVI")
    axes[1, 1].set_xlabel("log(1 + meters)")
    fig.suptitle("Phoenix key predictor and outcome distributions", fontsize=14)
    fig.tight_layout()
    output_path.parent.mkdir(parents=True, exist_ok=True)
    fig.savefig(output_path, dpi=200, bbox_inches="tight")
    plt.close(fig)


def _save_land_cover_figure(land_cover_df: pd.DataFrame, output_path: Path) -> None:
    top = land_cover_df.head(8).sort_values("share_pct")
    fig, ax = plt.subplots(figsize=(9, 5))
    ax.barh(top["land_cover_label"], top["share_pct"], color="#C56E3E")
    ax.set_xlabel("Share of Phoenix cells (%)")
    ax.set_title("Dominant Phoenix land-cover classes")
    ax.grid(axis="x", alpha=0.2)
    fig.tight_layout()
    output_path.parent.mkdir(parents=True, exist_ok=True)
    fig.savefig(output_path, dpi=200, bbox_inches="tight")
    plt.close(fig)


def _save_correlation_figure(corr_df: pd.DataFrame, output_path: Path) -> None:
    fig, ax = plt.subplots(figsize=(7, 6))
    image = ax.imshow(corr_df.values, cmap="coolwarm", vmin=-1, vmax=1)
    ax.set_xticks(range(len(corr_df.columns)))
```

```
    ax.set_yticks(range(len(corr_df.index)))
    ax.set_xticklabels(corr_df.columns, rotation=45, ha="right")
    ax.set_yticklabels(corr_df.index)
    for i in range(len(corr_df.index)):
        for j in range(len(corr_df.columns)):
            value = corr_df.iloc[i, j]
            ax.text(j, i, f"{value:.2f}", ha="center", va="center", fontsize=8, color="black")
    fig.colorbar(image, ax=ax, fraction=0.046, pad=0.04, label="Pearson correlation")
    ax.set_title("Correlation among key Phoenix variables")
    fig.tight_layout()
    output_path.parent.mkdir(parents=True, exist_ok=True)
    fig.savefig(output_path, dpi=200, bbox_inches="tight")
    plt.close(fig)

def _save_hotspot_map(gdf: gpd.GeoDataFrame, study_area: gpd.GeoDataFrame, output_path:
Path) -> None:
    plot_gdf = gdf.copy()
    plot_gdf["hotspot_10pct"] = plot_gdf["hotspot_10pct"].astype("boolean")
    centroids = plot_gdf.geometry.centroid
    background = gpd.GeoDataFrame(
        plot_gdf.loc[plot_gdf["hotspot_10pct"] != True, ["hotspot_10pct"]].copy(),
        geometry=centroids[plot_gdf["hotspot_10pct"] != True],
        crs=plot_gdf.crs,
    )
    hotspots = gpd.GeoDataFrame(
        plot_gdf.loc[plot_gdf["hotspot_10pct"] == True, ["hotspot_10pct"]].copy(),
        geometry=centroids[plot_gdf["hotspot_10pct"] == True],
        crs=plot_gdf.crs,
    )
    if len(background) > 80000:
        background = background.sample(n=80000, random_state=0)
    if len(hotspots) > 80000:
        hotspots = hotspots.sample(n=80000, random_state=0)

    fig, ax = plt.subplots(figsize=(8, 8))
    study_area.boundary.plot(ax=ax, color="black", linewidth=0.8)
    background.plot(ax=ax, color="#CFCFCF", markersize=1, alpha=0.35)
    hotspots.plot(ax=ax, color="#D64541", markersize=1.8, alpha=0.7)
    ax.set_title("Phoenix hotspot cells (top 10% LST)")
    ax.set_axis_off()
    fig.tight_layout()
    output_path.parent.mkdir(parents=True, exist_ok=True)
    fig.savefig(output_path, dpi=200, bbox_inches="tight")
    plt.close(fig)
```

Appendix B

GitHub Repository

The code used for data acquisition, preprocessing, feature assembly, and exploratory analysis for this project is maintained in a public GitHub repository. The repository includes scripts for constructing study areas and 30 m grids, assembling predictor layers, processing ECOSTRESS observations, and generating summary tables and figures used in this report, the specific code for which has been extracted into Appendix A. This repository is included for additional context not given in the simple functions defined to draw each figure. It should be noted as a disclaimer that this repository is still in active use and will be subject to change as we continue processing the remaining 29 non-Phoenix study cities.
Repository: https://github.com/goldember12-alt/urban-heat-mapping-dataset.